

KEMIONSAARI VANO, Finland

WMO: 027430

Lat: 59.869N

Lon: 22.193E

Elev: 8

StdP: 101.23

Time Zone: 2.00 (EUE)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-15.4	-12.8	-17.4	0.8	-15.2	-14.8	1.0	-12.0	15.4	-0.9	14.3	0.1	5.3	70	0.753

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.7	24.4	20.2	22.6	19.0	21.1	17.7	21.0	23.7	19.7	22.2	18.4	20.7	4.5	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.0	14.7	22.8	18.6	13.5	21.4	17.4	12.4	19.9	60.9	23.5	56.3	22.2	52.1	20.8	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.8	13.2	11.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-16.3	21.2	4.5	1.9	-19.5	22.6	-22.1	23.7	-24.7	24.8	-27.9	26.2	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.1	-2.3	-2.5	-0.4	3.7	9.6	14.0	18.0	17.4	13.6	7.9	4.5	1.2		
	DBStd	7.89	4.33	5.16	3.54	2.53	3.59	2.42	2.76	1.94	2.38	2.89	3.53	4.12		
	HDD10.0	1815	380	349	324	191	53	1	0	0	4	75	164	274		
	HDD18.3	4138	638	582	582	440	271	132	42	40	142	322	414	532		
	CDD10.0	762	0	0	0	1	41	121	247	231	112	11	0	0		
	CDD18.3	44	0	0	0	0	0	2	30	12	0	0	0	0		
	CDH23.3	106	0	0	0	0	0	2	93	11	0	0	0	0		
	CDH26.7	8	0	0	0	0	0	0	8	1	0	0	0	0		
(14) Wind	WSAvg	6.2	6.8	6.4	6.3	5.5	5.2	5.4	5.0	5.3	6.0	6.8	7.4	7.9		
(15) Precipitation	PrecAvg	657	58	42	38	33	31	46	61	74	61	74	78	63		
	PrecMax	795	113	93	77	57	61	85	147	153	148	140	158	107		
	PrecMin	493	9	3	6	7	9	12	2	9	18	35	6	21		
	PrecStd	77	26	24	17	15	15	19	37	36	33	29	37	26		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.2	4.4	6.6	12.7	21.0	22.9	27.7	24.6	19.3	14.2	10.5	8.5		
		MCWB	4.5	2.9	3.9	7.8	14.3	17.1	21.0	20.7	16.6	12.2	9.6	7.3		
	2%	DB	4.5	3.6	5.2	10.6	18.9	20.8	25.8	22.5	18.2	13.2	9.9	7.4		
		MCWB	3.8	2.7	3.3	6.8	13.5	15.9	20.7	19.1	15.6	11.9	9.3	6.4		
	5%	DB	3.7	3.1	4.3	8.7	17.2	19.1	24.0	21.2	17.4	12.4	9.2	6.6		
		MCWB	3.0	2.4	2.8	5.6	12.8	15.0	20.0	17.9	15.2	11.2	8.4	5.5		
	10%	DB	2.9	2.5	3.5	7.1	15.2	17.8	22.4	20.2	16.6	11.7	8.3	5.8		
		MCWB	2.2	1.9	2.3	4.8	11.7	14.4	19.2	17.2	14.8	10.4	7.4	4.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.0	3.6	4.6	8.3	15.3	18.4	22.5	21.7	17.3	13.0	10.1	7.7		
		MCDB	5.1	4.0	5.8	11.8	19.0	21.8	25.4	24.0	18.7	13.5	10.2	8.2		
	2%	WB	4.0	3.0	3.7	7.1	14.1	16.8	21.6	20.1	16.3	12.2	9.4	6.6		
		MCDB	4.4	3.4	4.8	10.1	17.8	19.5	24.4	22.1	17.5	12.9	9.7	7.1		
	5%	WB	3.2	2.5	3.1	6.1	13.2	15.8	20.7	18.7	15.7	11.6	8.6	5.7		
		MCDB	3.7	3.0	4.0	8.3	16.7	18.3	23.5	20.3	16.9	12.3	9.1	6.4		
	10%	WB	2.3	2.0	2.5	5.2	12.1	14.9	19.5	17.9	15.2	10.7	7.6	4.9		
		MCDB	2.8	2.5	3.4	6.7	15.0	17.3	22.2	19.7	16.3	11.5	8.3	5.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.7	3.5	4.2	4.6	5.4	4.9	4.7	4.0	3.5	3.0	2.6	3.0		
		MCDBR	2.8	2.7	3.8	6.9	8.3	7.1	6.7	5.2	4.2	3.0	2.6	2.9		
		MCWBR	3.0	2.4	2.8	4.2	4.3	3.7	3.2	2.8	2.9	2.8	2.8	3.1		
	5% WB	MCDBR	3.0	2.6	3.4	6.3	7.7	6.1	5.8	4.0	3.6	2.9	2.7	2.8		
		MCWBR	3.3	2.4	2.7	4.1	4.6	3.6	3.3	2.9	2.9	3.0	2.9	3.2		
(40) Clear-Sky Solar Irradiance	taub		0.284	0.317	0.335	0.373	0.368	0.362	0.391	0.383	0.353	0.337	0.312	N/A		
	taud		2.178	2.246	2.308	2.322	2.422	2.479	2.424	2.435	2.500	2.402	2.215	N/A		
	Ebn at Noon		486	677	793	825	860	871	832	810	778	657	441	N/A		
	Edh at Noon		43	70	89	105	103	99	102	94	75	61	42	N/A		
(44) All-Sky Solar Radiation	RadAvg		0.24	0.78	2.32	4.00	5.56	5.95	5.59	4.39	2.69	1.16	0.32	0.15		
	RadStd		0.04	0.14	0.27	0.41	0.53	0.39	0.48	0.41	0.32	0.16	0.06	0.02		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (56 neighbors)	+0.41	N/A	N/A	N/A	N/A	N/A	-133	N/A	N/A	N/A

Nomenclature: See separate page